

Predicting Employee Attrition

Causes, Model Performance, and Prevention Recommendations

This report summarizes an end-to-end analysis of employee attrition: exploratory data analysis of 1,470 employee records, a classification model that predicts who is likely to leave, and a SHAP-based explanation of what is driving those predictions. It closes with a set of concrete, prioritized recommendations for reducing attrition.

Dataset	1,470 employees · 3 departments
Overall attrition rate	25.3%
Primary model	Logistic Regression (class-balanced, standardized features)
Secondary model	Decision Tree (max depth 5)
Explainability	SHAP (Linear Explainer)
Tools used	Python (Pandas, Seaborn, scikit-learn, SHAP), Power BI

1. Exploratory Data Analysis

Attrition is not evenly distributed across the organization. The charts below break attrition down by department, salary band, promotion recency, overtime, job satisfaction, and tenure — the classic levers HR teams can actually influence.



Figure 1. Attrition rate across key organizational and compensation factors.

Key observations

- Human Resources has the highest departmental attrition rate (30.8%), though it is also the smallest department by headcount, so absolute numbers are modest.
- Employees working overtime leave at nearly double the rate of those who don't (38.0% vs. 20.4%) — the single largest gap in the dataset.
- Attrition falls steadily as monthly income rises: employees in the lowest salary band leave at roughly triple the rate of the top band.
- Job satisfaction is inversely related to attrition: the lowest satisfaction group (level 1) attrites at 31.4%, versus 22.2% at the highest level.
- Attrition is highest in the first 2-3 years of tenure and declines the longer someone stays — the first two years are the highest-risk retention window.

Correlation structure

A correlation heatmap of all numeric features confirms the same story quantitatively: monthly income, total working years, job level, and age are the strongest negative correlates of attrition (i.e., higher values associate with staying), while distance from home and training frequency show weak positive association.

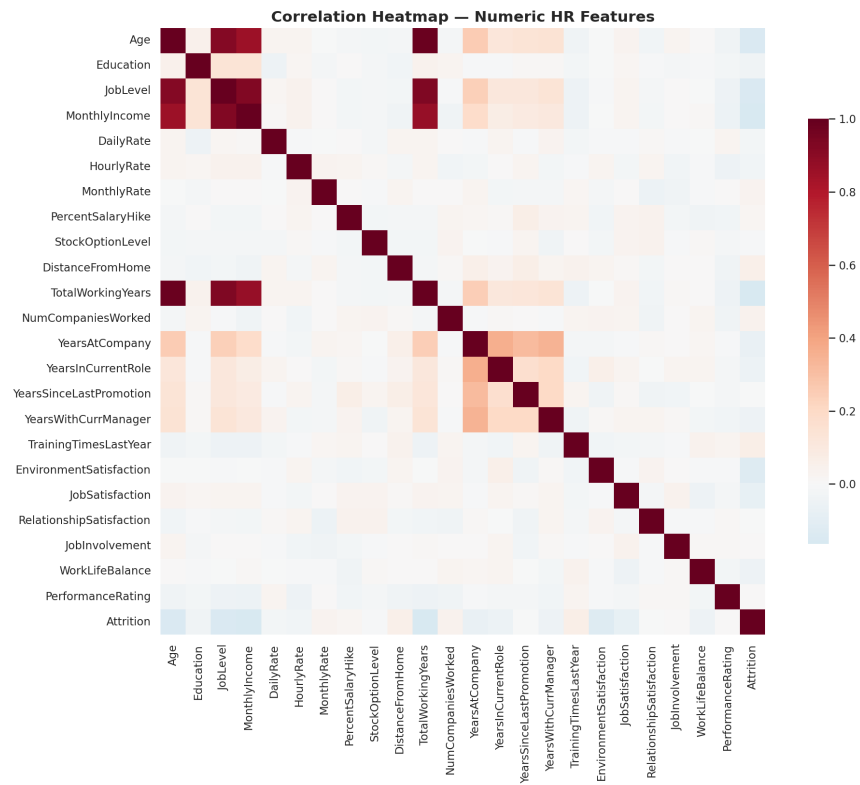


Figure 2. Correlation heatmap of numeric HR features (including Attrition as 0/1).

2. Classification Model & Accuracy Report

Two classifiers were trained to predict attrition from 41 engineered features (demographics, compensation, satisfaction scores, tenure, and role): a Logistic Regression model and a Decision Tree (max depth 5, for interpretability). Both use class-balanced weighting because only 25% of employees in the dataset actually left — without balancing, a model could reach 75% accuracy by never predicting attrition at all, which would be useless for HR.

Metric	Logistic Regression	Decision Tree
Accuracy	69.0%	60.2%
Precision (Yes)	42.7%	34.5%
Recall (Yes)	67.6%	64.9%
F1-score (Yes)	52.4%	45.1%
ROC AUC	0.732	0.637

The Logistic Regression model was selected as the primary model: it achieves higher accuracy (69.0%) and a notably higher ROC AUC (0.73 vs. 0.64), meaning it ranks at-risk employees more reliably. Recall on the "Yes" (attrition) class was prioritized over raw accuracy, since for a retention program, missing a departing employee (a false negative) is more costly than flagging someone who ultimately stays.

Confusion matrices

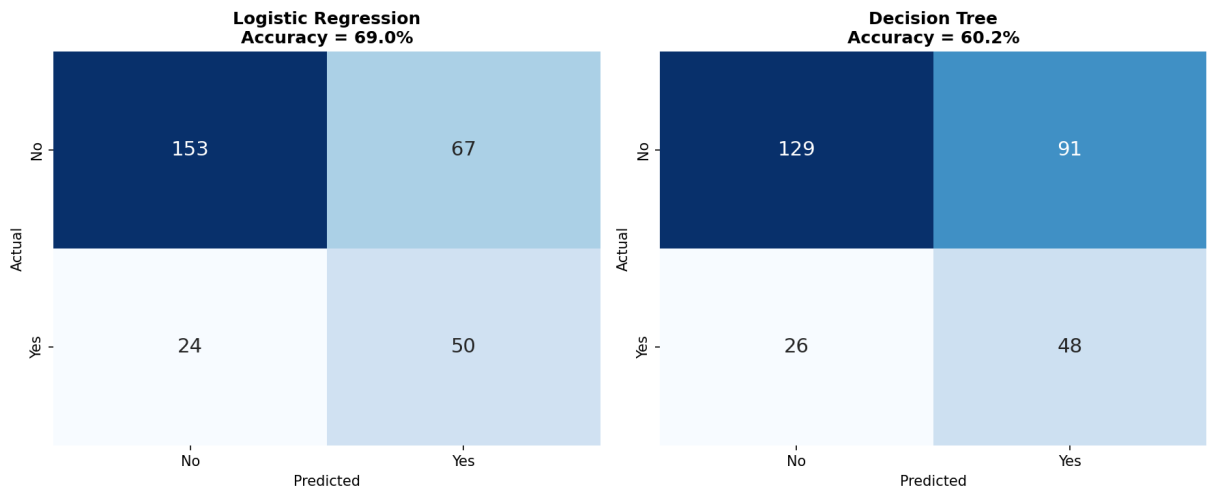


Figure 3. Confusion matrices on the held-out 20% test set (294 employees).

ROC curve

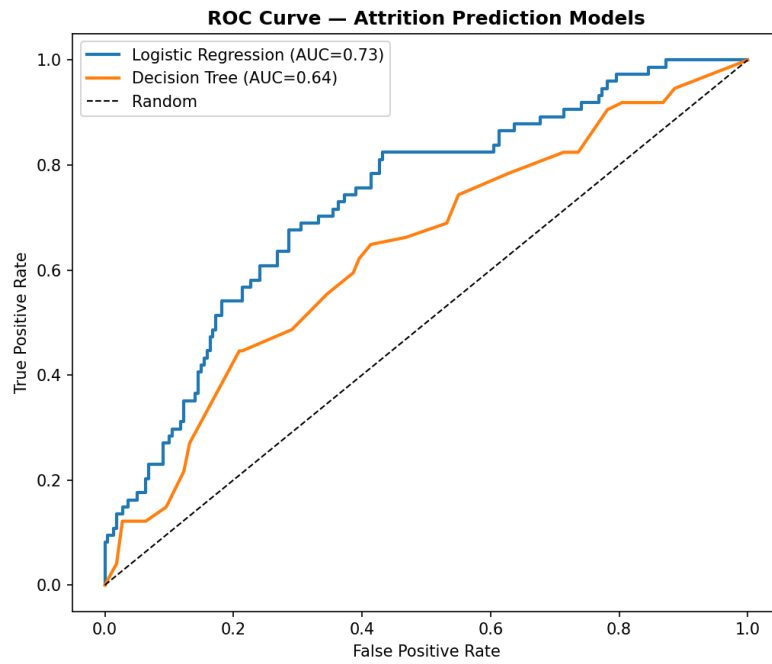


Figure 4. ROC curves comparing both models against a random baseline.

3. SHAP Value Analysis

SHAP (SHapley Additive exPlanations) values quantify how much each feature pushes an individual prediction up or down, relative to the model's average output. This makes the model auditable: HR can see not just who is at risk, but why.

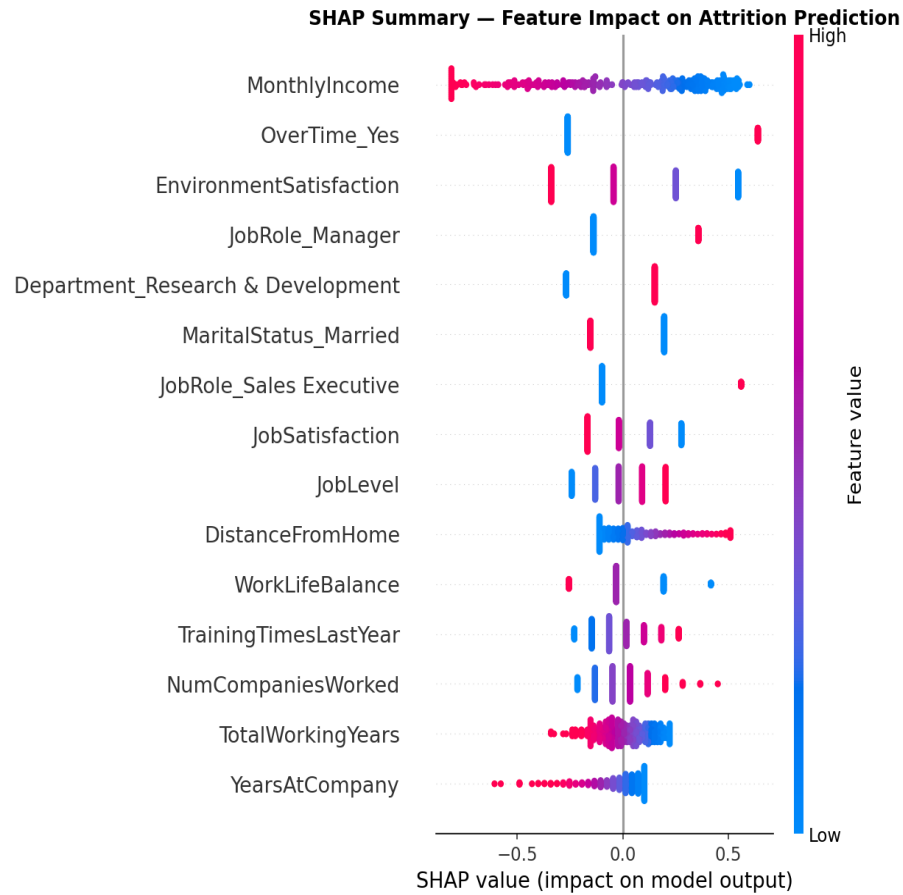


Figure 5. SHAP summary plot. Each dot is one employee; red = high feature value, blue = low. Position left/right shows whether that value pushed the prediction toward staying or leaving.

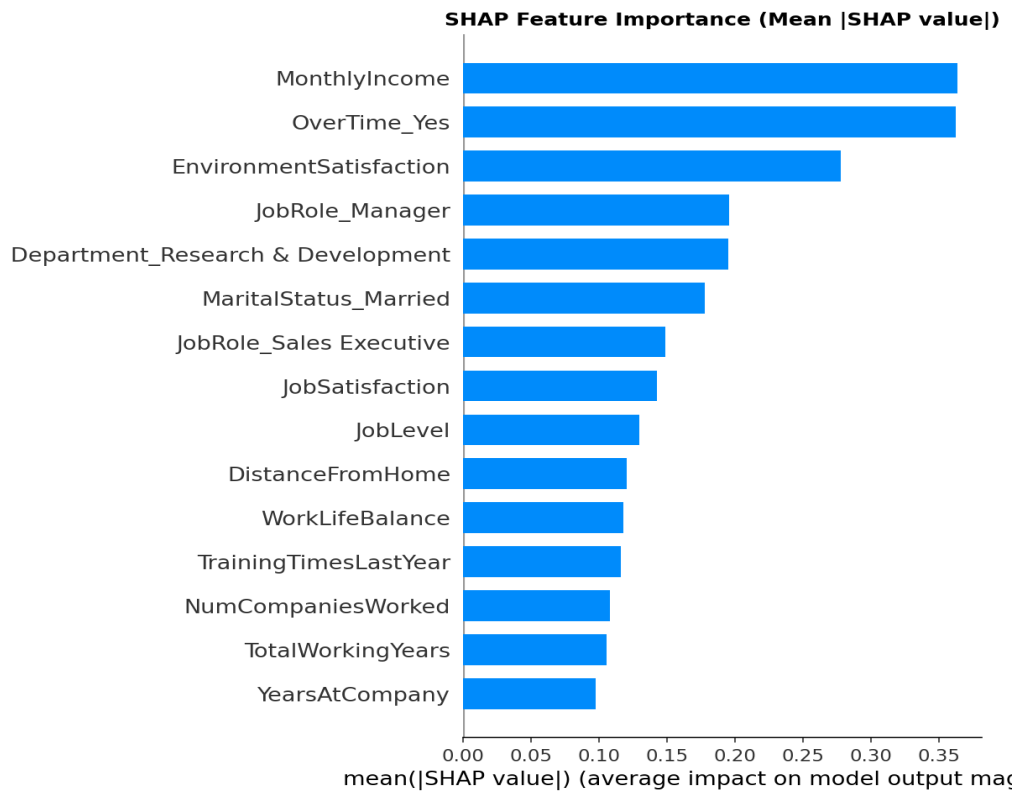


Figure 6. Mean absolute SHAP value per feature — overall importance ranking.

What the model is actually keying on

- MonthlyIncome and OverTime are the two dominant drivers — low pay combined with overtime work is the strongest attrition signal the model finds.
- EnvironmentSatisfaction is the third-strongest factor: employees who rate their environment poorly are consistently pushed toward the 'leaving' prediction.
- Being in a Manager or Sales Executive role, and working in R&D, shift risk upward slightly once income and satisfaction are accounted for.
- Married employees are somewhat less likely to leave than single or divorced employees, holding other factors constant.
- Work-life balance and job satisfaction contribute meaningfully but are smaller than income and overtime — they matter most as tie-breakers for employees already near the decision boundary.

4. Attrition Prevention Recommendations

Recommendations below are ordered by expected impact, based directly on the SHAP-ranked drivers and EDA patterns above. Each pairs a diagnosis with a concrete action.

1. Audit and cap chronic overtime

Overtime is the single strongest behavioral driver of attrition (38.0% vs. 20.4% attrition rate). Identify teams and roles with sustained overtime, investigate root causes (understaffing, unrealistic deadlines, poor workload distribution), and set a policy trigger — e.g., a manager review — when an employee logs overtime in 3+ consecutive review periods.

2. Address pay compression in the lowest income band

Employees earning under \$3,000/month leave at roughly 3x the rate of the top earners, and income is the top SHAP driver overall. Benchmark the bottom salary band against market rates and prioritize corrective raises or promotion pathways for employees who are both underpaid and tenured (a strong flight-risk combination).

3. Fix the first-two-years onboarding and engagement gap

Attrition peaks in years 0-3 of tenure and declines steadily after. Strengthen onboarding, assign mentors for the first 12 months, and schedule structured check-ins at 30/90/180 days to catch dissatisfaction before it becomes a resignation.

4. Treat low environment/job satisfaction scores as an early-warning signal

Both satisfaction metrics rank in the model's top drivers. Route employees who score 1-2 (of 4) on either metric in engagement surveys to a manager or HR conversation within a set window, rather than waiting for an annual review cycle.

5. Build a promotion cadence for stalled careers

Employees who haven't been promoted in 5+ years show elevated attrition. Implement a periodic career-pathing review for anyone that far past their last promotion, even if a promotion isn't the final outcome — a documented growth conversation itself reduces flight risk.

6. Prioritize retention outreach using the model's risk tiers

Use the PredictedAttritionRisk score (exported alongside this report) to focus limited HR time on the highest-value conversations first: start with 'Critical' and 'High' tier employees who are also high performers or hard to replace, rather than spreading retention effort evenly across the workforce.

7. Watch Manager and Sales Executive roles, and the R&D department, specifically

These segments carry elevated risk in the model even after controlling for income and satisfaction, suggesting role-specific stressors (span of control, quota pressure, project churn) worth a targeted pulse survey.

Note: this analysis uses a synthetically generated dataset structured to reflect realistic HR attrition patterns (based on the well-known IBM HR Analytics schema), built for demonstrating the analytics workflow end-to-end. Before acting on these recommendations operationally, re-run this pipeline against your organization's actual HR data — the modeling code (Python), Power BI dataset, and dashboard included with this deliverable are built to accept a real HR export with the same column structure.